HANYU YANG

CRM DATA ANALYTICS / SOFTWARE SYSTEMS / APPLIED AI

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PROFILE

Software engineering graduate and postgraduate student in AI, Ethics and Society at the University of Hong Kong. Experience spans CRM data analytics in financial services, AI workflow development, full-stack products, multimodal research, and distributed data systems. I work best where technical evidence, business context, and clear delivery meet.

01 / EXPERIENCE

Professional Experience

HSBC - Wealth and Personal Banking
Mar. 2026 - Jul. 2026
CRM Data Analyst Intern
Guangzhou, China

- Integrated core-banking, credit-card, and mobile-channel customer data with SQL and Python on GCP BigQuery and Data Science Workbench, supporting a unified customer view and precise WPB profiling.
- Performed data cleaning, exploratory analysis, and predictive modeling in Python to support customer segmentation and evidence-led marketing decisions.
- Contributed to precision-marketing strategy and campaign deployment through SmartCRM and PEGA, supporting omnichannel customer lifecycle management.
- Supported omnichannel customer-communications delivery by coordinating handover requirements, UAT inputs, execution readiness, channel schedules, and stakeholder sign-offs across in-app, eDM, and marketing-push journeys.

TOOLS SQL / Python / GCP BigQuery / Data Science Workbench / SmartCRM / PEGA / UAT

Guangzhou Xin'an Data Co., Ltd.
Jul. 2025 - Aug. 2025
AI Development Intern - Digital Innovation Department
Guangzhou, China

- Designed and deployed a full-stack AI prompt-optimization platform with authentication, community libraries, and knowledge-sharing capabilities for more than 100 internal users.
- Integrated Model Context Protocol with Dify to automate AI-driven reporting workflows.
- Developed RESTful APIs for third-party integration and enterprise data exchange.

TOOLS MCP / Dify / REST API / Full-stack systems / AI workflows

02 / EDUCATION

Education

Master of Arts in AI, Ethics and Society
Sept. 2026 - Aug. 2027
The University of Hong Kong - Hong Kong SAR

Postgraduate study connecting AI systems with their ethical and societal contexts.

Bachelor of Science in Software Engineering
Sept. 2022 - Jun. 2026
Guangdong University of Foreign Studies - School of Information Science and Technology

GPA 88/100 - Top 10%. Selected modules: Text Information Processing (99), Network Programming (96), Natural Language Processing (96), Software Architecture (96), Big Data Processing Technology (94), Further Mathematics (91), and Web Design (90).

03 / RESEARCH

Research Experience

Guangdong University of Foreign Studies — **Oct. 2024 - Dec. 2024**, Guangzhou, China

Research Contributor - Fake News Detection through Multimodal Attention

- Proposed a semantic-conflict recognition and multimodal-attention framework for image-text inconsistency.
- Added KL-divergence and MSE modules to CLIP alignment for quantifying cross-modal semantic discrepancies.
- Combined BERT, ResNet-50, and cross-modal self-attention, reaching 88.9% accuracy and improving on the CLIP baseline by 2.8%.

TOOLS BERT / ResNet-50 / CLIP / Multimodal attention

University of Cambridge — **Aug. 2024**, Cambridge, UK

Research Leader - Skin Cancer Classification with CNN and Attention

- Led development of a lightweight enhanced-AlexNet CNN for benign-versus-malignant dermoscopic lesion classification.
- Integrated spatial attention to focus the model on diagnostically relevant regions.
- Improved classification accuracy by 15% through preprocessing optimization and background-noise suppression.

TOOLS CNN / AlexNet / Spatial attention / Medical imaging

Guangdong University of Foreign Studies — **Oct. 2023 - Nov. 2023**, Guangzhou, China

Algorithm Designer - Autonomous Parking Detection

- Redesigned YOLOv5s with a GhostNet backbone and CBAM attention for parking-slot detection under occlusion and variable lighting.
- Trained in PyTorch with RMSProp and Mosaic augmentation for 500 epochs on RTX 3090.
- Reached 97.1% AP and 93.7% AR while reducing inference latency by 20% against the YOLOv5s baseline.

TOOLS PyTorch / YOLOv5s / GhostNet / CBAM

04 / ENGINEERING

Selected Engineering Projects

International Chinese Education Platform - Front-End Developer — **Apr. 2025 - Jul. 2025**

Vue 3 / Composition API / Real-time UX

Built a responsive student portal with real-time class management and messaging for more than 500 international students. Added lazy loading and virtual scrolling for smooth interaction under real-time data flow.

YouTube English Comment Sentiment Analysis - Project Leader — **May. 2025 - Jun. 2025**

NLTK / TF-IDF / Word2Vec / LinearSVC / XGBoost / Streamlit

Collected more than 50,000 comments across five topics; built and tuned classical-ML feature pipelines. Reported 90.5% validation accuracy and 0.89 F1; deployed a sub-two-second Streamlit interface for 100+ internal users.

Global Economic Data Analysis - Data Engineer — **Nov. 2024 - Dec. 2024**

Hadoop / Hive / Spark / Flink / Kafka

Processed 8,519 records with 27 features on a three-node Hadoop cluster. Built batch and streaming pipelines across 150+ countries; reduced complex batch-query response time by 40% and improved time-series queries by 15%.

ChatGDUFS Campus Q&A - NLP Developer — **Nov. 2024 - Dec. 2024**

LangChain / RAG / Chroma / Qwen-7B / LoRA / Flask

Built a RAG pipeline with 2,000+ document embeddings and a sub-two-second Flask query API. Fine-tuned Qwen-7B with LoRA on 5,000 campus Q&A pairs, with reported domain accuracy improving from 72% to 86%.

05 / CAPABILITY

Technical Toolkit

PROGRAMMING	FRAMEWORKS & SYSTEMS	LANGUAGES
C++ / Java / Python / SQL / Vue 3 / React / JavaScript / Git	TensorFlow / PyTorch / MySQL / Docker / Redis / Hadoop / Spark / Flink / Kubernetes	English - IELTS 7 Mandarin - Native Cantonese - Fluent

06 / RECOGNITION

Awards & Recognition

01	China International College Students' Innovation Competition - Outstanding Award, Jul. 2025.
02	14th Challenge Cup Guangdong College Students Entrepreneurship Competition - Bronze Award, Jul. 2024.
03	National College Students English Vocabulary Ability Competition - Third Prize (National), Jun. 2024.
04	Guangdong University of Foreign Studies - Third-Class Scholarship for Outstanding Students, Oct. 2023.
05	Guangdong Youth Science and Technology Innovation Competition - Silver Award, Oct. 2019.

07 / PUBLICATION

Publication

Yang, H., Bai, J., He, C., Wang, Y., & Liang, W. (2024). Synthesis, characterization, and testing of rare earth-doped GdBO3 luminescent materials. Guangdong Chemical Industry, 51(10), 50-54.

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